

END TERM EXAMINATION

THIRD SEMESTER [B.TECH]] FEBRUARY 2023

Paper Code: ECC-207

Subject: Digital Logic And Computer Design

Time: 3 Hours

Maximum Marks: 75

**Note: Attempt five questions Q.No. 1 which is compulsory.
Select one question from each Unit. Internal choice is indicated.
Assume missing data, if any.**

Q1. Attempt all questions: (3x5=15)

- (a) Write the base of the following number systems: Decimal, Binary, Octal, and Hexadecimal.
- (b) Draw symbol and write the truth table of JK flip flop.
- (c) State the necessity of multiplexer.
- (d) Write about parallel priority interrupts.
- (e) List out the typical characteristic of multiprocessors.

UNIT-I

- Q2. (a) State and prove De Morgan's Theorems. (5)
(b) Design 1: 16 demultiplexer using 1: 4 demultiplexers. (10)
- Q3. (a) Draw the circuit diagram of BCD to 7 segment decoder and write its truth table. (7)
(b) Simplify the following Boolean function,
 $f(W,X,Y,Z) = \sum m(2,6,8,9,10,11,14,15)$
Using Quine-McClukey tabular method. (8)

UNIT-II

- Q4. (a) Describe the working of Master-Slave JK Flip-Flop with Truth Table and Logic diagram. (7)
(b) Describe the operation of 4 bit SISO shift register with the help of block diagram, truth table and timing diagram. (8)
- Q5. (a) Draw the block diagram of Programmable Logic Array. (7)
(b) Define modulus of a counter? Write down the number of flip flops required for mod-5 counter? (8)

UNIT-III

- Q6. (a) Explain the organizations of micro programmed control unit with neat sketch. (8)
(b) What are the different phases of a basic computer instruction cycle? Explain instruction cycle with flowchart. (7)
- Q7. (a) Explain with a neat diagram, system configuration incorporating an I/O processor. (8)
(b) Discuss the following: Computer configuration for micro program, Symbolic micro program and binary micro program. (7)

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UNIT-IV

- Q8. (a) Show internal configuration of a DMA controller diagrammatically and explain how it's working. (8)
 (b) Explain Types of Interrupts with an example for each. (7)
- Q9. (a) Explain how memory management unit provides memory protection. (7)
 (b) Explain Cache with Set-Associative and direct mapping. Assume your own example address and explain. (8)
